

Section 2: System Design Fundamentals

Understanding IP Addresses

Imagine trying to send a letter without an address. The postal service wouldn't know where to deliver it and your letter would never reach its destination. The same principle applies to the internet. Every device needs an address to send and receive data. This is where an IP address comes into picture.

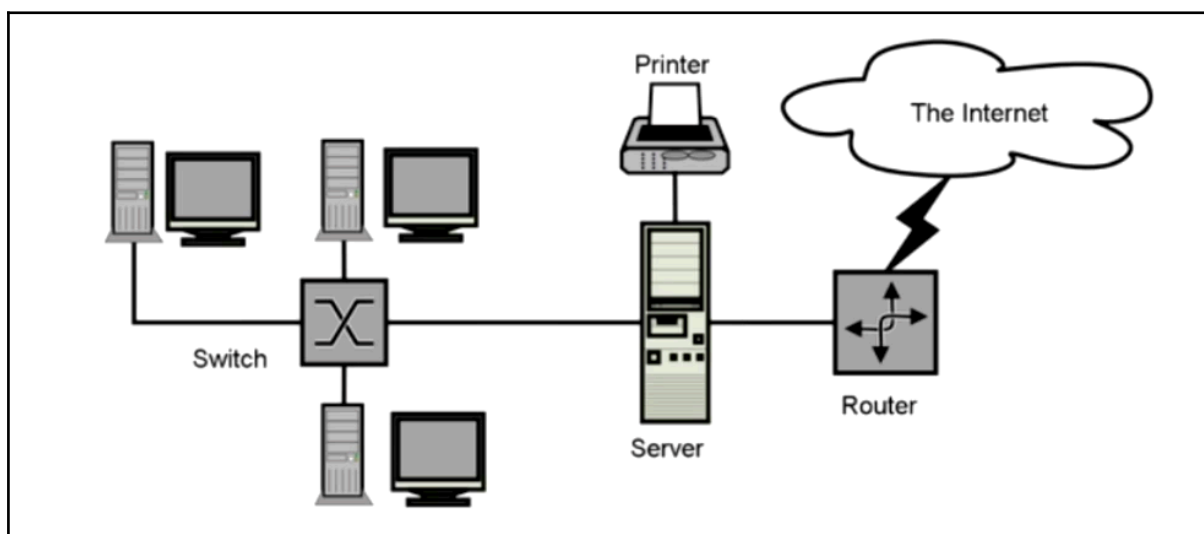
Whether we are designing a distributed system, working with cloud infrastructure or preparing for system design interviews, understanding IP addressing is essential. Every online service, be it Netflix, Amazon, or even a simple website, relies on a well structured IP management system to ensure seamless communication and scalability.

In this section we are going to break down one of the most fundamental networking concepts, which is IP addresses. We will be discussing:

- What is an IP Address and why is it so important in networking?
- How do IP addresses work?
- Difference between IPv4 and IPv6
- Role of private and public IPs in system architecture.

Introduction to IP Addresses

Every time we open a website, send an email, or stream a video, your device is communicating with other machines across the internet. But how does it know where to send this request? The answer lies in IP addresses.



An IP address, or Internet Protocol Address, is a unique identifier assigned to every device connected to the internet. It's just like a home address, where a mail carrier can use it to know exactly where to deliver the mail to. Your computer also has an IP address of its own, which allows computers, servers and other devices to locate and communicate with each other when they are connected to the internet.

IP addresses are very important because:

- Without IP addresses, there would be no way to send or receive data online. They are the foundation of all networking.
- They enable devices to exchange information seamlessly.

There are mainly 2 versions of IP addresses:

- IPv4
- IPv6

And they can be classified further as public IPs and private IPs.

What is IPv4?

IPv4 (Internet Protocol Version 4) is the most commonly used version of IP addressing. It was actually created a few decades ago when people were actually designing a solution for giving the addresses to every computer on the network. It uses a 32 bit addressing scheme, which means that it can support approximately 4.3 billion unique addresses.

Now when the internet was first designed, no one expected the number of connected devices to explode as it has today. So the original IP addressing scheme, the IPv4, so far has been the backbone of the internet and it has been for decades.

However, since there is a limitation of only supporting 4.3 billion unique addresses, it is actually falling short. Despite this limitation, IPv4 is still widely used because it is deeply embedded in the networking infrastructure, cloud services and corporate networks. But due to the rapid growth of the internet, IPv4 addresses are running out and eventually led to the development of IPv6.

The overall structure of an IPv4 address is typically written in a dotted decimal format. It consists of 4 groups of numbers separated by dots and each group can have numbers ranging from 0 to 255, and these are actually called octets. Every octet has a specific responsibility, like the first one represents the class of the network, the second one and the third one represent what subnet it belongs to and the fourth one represents what actual device is on the network that it is identifying.

192.168.1.1

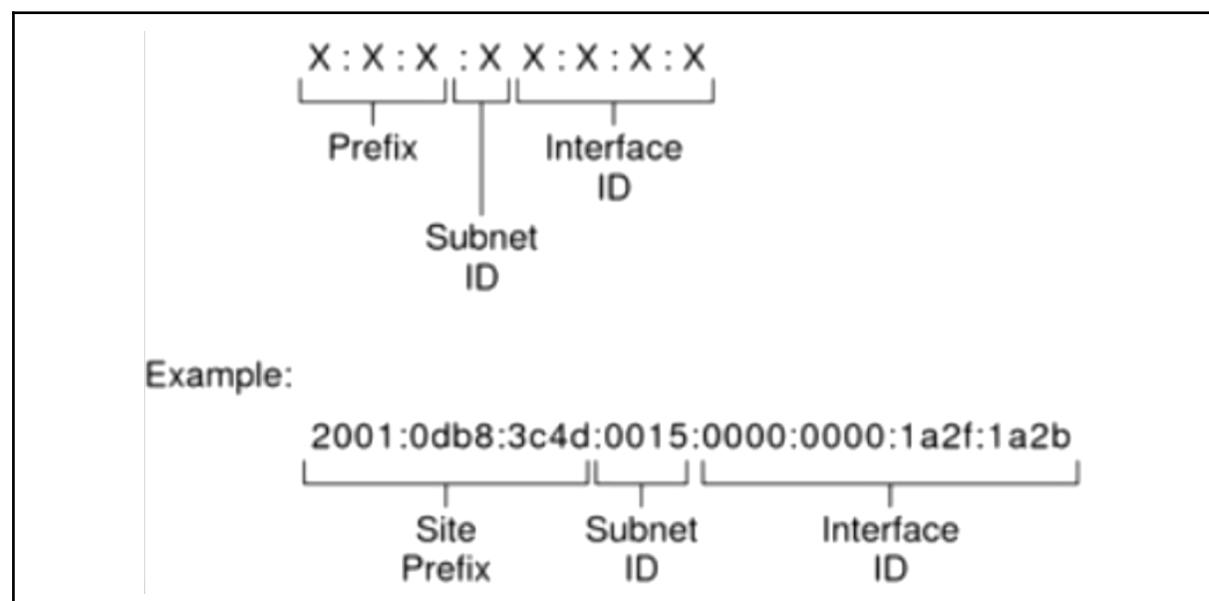
This is what an IPv4 address looks like, and it can support up to 4.3 billion unique devices.

What is IPv6?

IPv6 is the successor of IPv4, which solves the address shortage problem of IPv4. In today's context, imagine trying to fit the entire internet into just 4.3 billion addresses. Achieving this is impossible using the IPv4, this is exactly why IPv6 was introduced.

IPv6 (Internet Protocol Version 6) is the next generation IP addressing system, which is designed to replace the IPv4. It uses a 128 bit address instead of 32 bit address, which allows for an almost infinite number of unique addresses, (3.4×10^{38}) 340 undecillion to be exact.

Unlike IPv4, which was written in dotted decimal format, IPv6 addresses used hexadecimal notations, which are separated by colons, as you can see here in the diagram.



Each IPv6 consists of 8 groups of 4 hexadecimal digits, making it much longer than the IPv4. This format of IPv6, provides enough addresses for every device on this earth to have a unique address of their own.

IPv6 isn't just about more addresses, it is also bringing some key improvements in the IP addressing scheme. Improvements like:

- Better Performance and Security
- Simplified Network Management
- Faster Data Processing
- Efficient Routing and Packet Handling

With the growth of cloud computing devices, IoT devices and global internet expansion, IPv6 adoption is actually increasing at a rapid rate. You can look up at the above diagrams, we can actually see how IPv6 is different from IPv4. It shows IPv6 has larger address space, which eliminates the need for address translations from private addresses to public IP addresses.

What is Network Address Translation (NAT) ?

To understand Network Address Translation(NAT), we need to first understand other critical concepts of public IPs and private IPs, and try to understand what their role is in the system design.

Think of the entire internet as a big city, your home as a street address and identifies it to the world. This street address is like the public IP address. But inside your home, your device has its own local addresses, like apartment numbers, which is sort of like a private IP address.

A public IP address is assigned to a device by an internet service provider, that is globally unique and it allows the devices to communicate directly over the internet. The example of public IP addresses includes the IP addresses that are assigned to websites, cloud servers, and the router that is actually in your home or in any business, whatever router that is actually connected facing the internet.

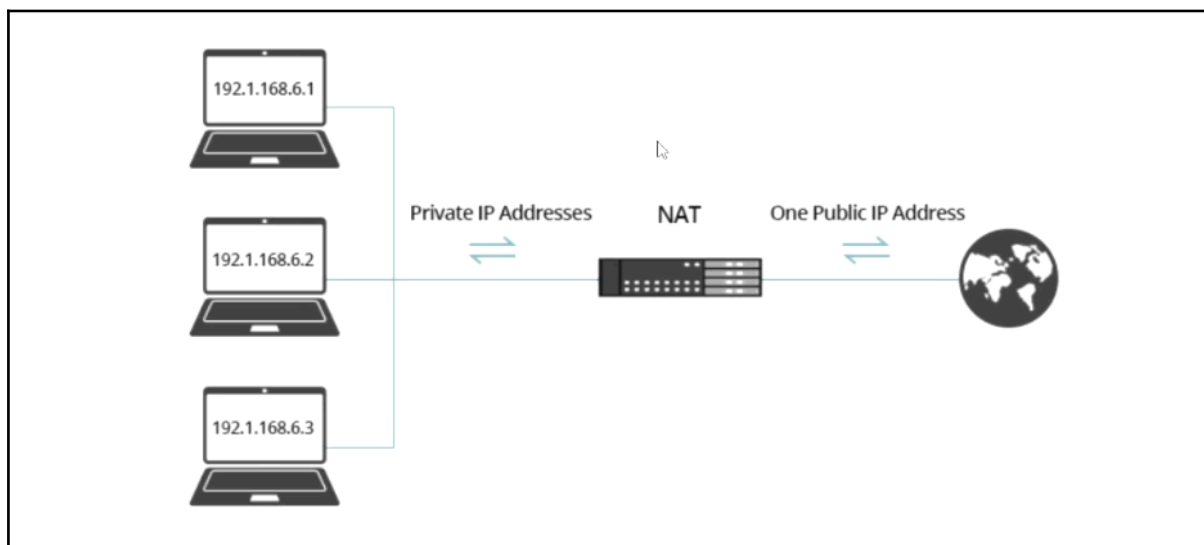
On the contrary, the private IP address is used within a local network and it is not accessible from the internet. These addresses have devices inside a local network, inside the same network to identify each other. Like in your home network, you might have laptops, smart TVs, phones for all these devices to communicate with each other. All of them will have a private IP address. You'll often see these private IP addresses in homes and office networks because private IPs are reusable and they don't require global registration.

Because private IPs are reserved and reusable, multiple homes and offices can use the same private IP addresses without any conflict. Another home might have a different public IP address (for example, 203.45.67.89), but inside that home, they can use the exact same private IP addresses as your home, such as 192.168.6.1 for the router or 192.168.6.10 for a laptop.

This is possible because private IP traffic never leaves the local network; the router uses a process called Network Address Translation (NAT) to convert private IPs into the public IP when communicating with the internet.

Understanding the difference between private IPs and public IPs is very crucial for designing scalable and secure systems, due to:

- **Security:** Private IPs isolate the internal system from external threats. So even if anyone gets hold of your private IP address, since this device is not accessible from outside, they will not be able to access this device just by knowing their private IP address.
- **Scalability:** It allows organizations to use many internal addresses, without needing the public IP addresses.
- **Network Management:** It helps to manage the cloud networks, virtual private networks and enterprise systems much more efficiently.



Look at the above diagram, which shows a typical network. The devices inside a home, which are on the left side, they use all private IP addresses to communicate with each other, and whenever they want to communicate to the outside world, they give their private addresses as the referrer to the outside world, while the router itself has a public IP address that is provided by your ISP.

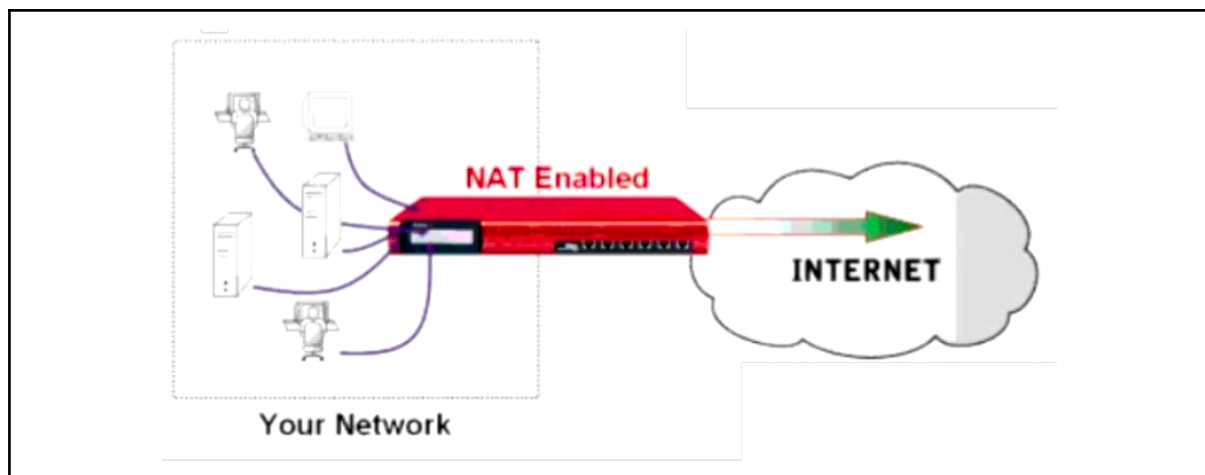
Why need Private IPs?

Imagine if every device in our home, your phone, laptop, smart TV and even your smart fridge needed a unique, globally accessible IP address, meaning public IP address. Now with billions of devices that we have worldwide right now, we would run out of IP addresses almost instantly like that. We need private IPs to avoid letting that happen.

There are 4 main reasons for having private IP addresses.

- **Address Conservation:** IPv4 has a limited number of addresses and using private IPs allow organizations and homes to reuse the same IP ranges without exhausting the global address pool.
- **Enhanced Security:** Private IP addresses are not directly reachable from the internet, making them much more secure. This fact prevents direct external attacks on devices that are inside our network.
- **Efficient Network Management:** Businesses and homes can assign private addresses to multiple devices and manage their internal networks efficiently without sending a separate public IP for each device.
- **Cost Effective:** Using the private IPs reduces the need to purchase multiple public IP addresses, which sometimes can be very expensive, especially when it comes to large enterprises.

Private IPs are used inside corporate networks, cloud infrastructures, and data centers, and they ensure smooth communication between internal services while limiting exposure to the public internet.

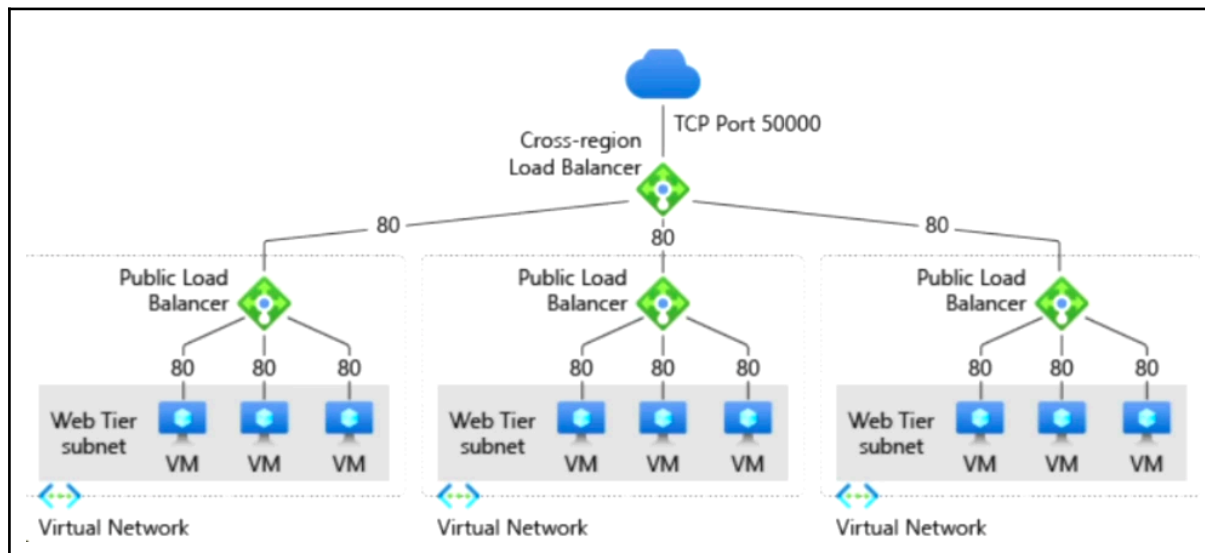


The above diagram shows a typical home network where multiple devices which are inside the home use private IPs, while a single public IP is assigned to the router for all the external communication needs.

The Role of IPs in System Design

This section shows a detailed version of how IP addresses are important in system design.

- **Scalability:** We can use any number of IP addresses within a region or within a private network when we are actually hosting our applications on a cloud platform or on our on-premises servers.
- **Security:** It is also much more secure because we enable firewall rules, create virtual private networks, and we can have private networks within our hosting infrastructure to have better security.
- **Load Balancing:** The load balancer can actually efficiently do load balancing within the network using all the private IP addresses.
- **Cloud Networking:** The use of private IP addresses is actually very exhaustive, it helps to design the system architecture that is leveraging the private IPs, when designing the large scale systems.



Common System Design Interview Questions on IPs

- *How do IPv4 and IPv6 addresses differ?*
- *Why do we need private IPs in system design?*
- *How does NAT help in addressing the IPv4 shortage?*
- *Explain how a load balancer distributes traffic using IPs.*
- *How does DNS resolve IP addresses in large-scale systems?*